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Signal processing and efficient ML under constraint

I work on systems where being confidently wrong is the expensive failure. That has taken me through motion artifact rejection in wearable photoplethysmography, multimodal classification under a small label budget, deterministic matching with selective learned fallback, and a specification for spectral imaging built around refusing to render what the sensor did not measure. Fourteen projects, four of them paused or retired with the reason recorded.

Research interests

Signal recovery under motion and noise, and the design of trust gates that discard corrupted measurements rather than smoothing over them. Cost efficient inference: when a learned component earns its compute against a deterministic baseline, and how to measure that honestly. Uncertainty quantification and provenance in generated and reconstructed outputs, including spectral reconstruction where the failure mode on metamers is known and must be stated. Hardware software co-design for embedded sensing under power, thermal and dimensional constraint.

Education

B.Tech, Computer Science Engineering (Data Science) Aug 2023 – Aug 2027

Swami Vivekananda Institute of Technology, Hyderabad

- CGPA 7.85 / 10. Class rank in the top 25% of 120 students.
- Coursework: machine learning, artificial intelligence, computer vision, natural language processing, data structures and algorithms, database systems, probability and statistics.

Research projects

OptiXon: continuous physiological sensing in prescription eyewear 2024 – Jul 2026

Founder and technical lead. Second place, Techno Udbhav 2025 inter college research competition.

- Research question: can a prescription eyeglass frame carry photoplethysmography and inertial sensing for all day physiological monitoring, inside a 138 × 10.5mm temple arm, without a camera, a display, or a battery that needs charging at midday.
- Signal approach: the sensor sits at the temple tip over the superficial temporal artery. The dominant problem is not sensor selection but motion artifact, since walking and head rotation corrupt the waveform while leaving it smooth and plottable. The gyroscope is used as a trust gate that flags corrupted windows before processing, so the system discards rather than averages.
- Power: a written power budget preceded component selection. Moving from ESP32 to the nRF52 family cut projected draw by roughly 82%, which is the difference between charging at lunch and charging overnight. Duty cycled sampling yields a projected 18 to 24 hours on a 200mAh curved cell.
- Verification: board, enclosure and routing are generated by Python against KiCad and FreeCAD, so every change is re-run and re-verified rather than redrawn. Electrical rule check clean, netlist machine verified, protection circuit confirmed in ngspice, firmware compiles and links against the nRF5 SDK.
- Safety finding: the battery protection pairing carried a pin mapping error that passed visual review. The circuit would have charged, discharged and behaved normally in every functional test while providing no overcurrent or overdischarge protection on a lithium cell worn against the head. Found by verifying the same mapping three independent ways and logged as findings 37 and 38 of 51.
- Status and honesty: the paper phase is complete and no physical prototype exists. Battery life, thermal margin and ingress protection are design estimates and are labelled as estimates wherever they appear.
- External review: the design and workflow were shown to M.Tech students at IIT Hyderabad across multiple sessions, and to a working hardware professional on microcontroller architecture and sensor selection. Both confirmed feasibility and suggested changes. There was no academic supervision on this project, so every design, simulation and firmware decision is my own.

`nRF52832, MAX30102, LSM6DSOX, KiCad, FreeCAD, ngspice, nRF5 SDK, C, Python`

Multimodal fusion for classification under a small label budget

2024, rebuilt Feb – Apr 2026

- Problem: twenty visually overlapping classes from 1,297 labelled images, where receipts, plain quote cards and bank statements are all dark text on a light rectangle and are separable only by what the text says.
- Version one, a fine tuned convolutional network on roughly 12,000 scraped images with K-means proposing groupings before labelling, plateaued. Additional data did not move it.
- Diagnosis came from the confusion matrix rather than from the loss curve: errors concentrated in text bearing classes, which indicates a missing input channel rather than insufficient capacity.
- Version two concatenates an EfficientNet-B3 embedding with a 500 term TF-IDF vector over extracted text and classifies with logistic regression. 96.92% accuracy on a 260 image held-out set, nine classes at perfect precision and recall.
- Methodological note: joining a dense embedding to a sparse lexical vector lets the dense half dominate on scale alone, so the fusion was treated as a modelling decision. The head was kept linear so each modality's contribution stayed inspectable, and the two weakest classes were traced to the same cause, images carrying no readable text.
- The version one figure was training accuracy and the version two figure is a held-out test set. They are not comparable and are not presented as if they were.

`PyTorch, EfficientNet-B3, OCR, TF-IDF, scikit-learn`

BeyondVision: provenance constrained spectral imaging

2026

- A specification for spectral capture on consumer hardware, organised around a single rule: every pixel is tagged MEASURED, RECONSTRUCTED or SIMULATED, and nothing exports without its tag.
- The physics document was written first and establishes hard limits before any design. Silicon's 1.12 electronvolt bandgap places an absolute wall at 1107nm, which makes thermal imaging from a phone camera physically impossible rather than merely difficult. Writing this document eliminated four proposed features before any implementation.
- Five hardware tiers are specified from zero cost to roughly 250 dollars, each with its honest provenance: an infrared pass path giving measured 700 to 1000nm, a filter removed sensor giving measured 350 to 1000nm, a narrowband LED array producing a genuine 24 band measured spectral cube for around 150 dollars, and a microbolometer for measured 8 to 14 micrometres.
- Reconstruction from three broadband channels is treated as inference with a quantified error and a known failure mode on metamers, and the specification states that failure rather than concealing it.
- Radiometric integrity requires beginning at RAW, since a consumer image pipeline applies six or more undocumented non-linear scene dependent transforms before a JPEG exists.
- Status: specification only. No implementation code exists, and the document set says so.

`Optics, spectral reconstruction, radiometric calibration, sensor characterisation`

Deterministic matching with selective learned fallback

May 2025 – present

- Studies where a learned matcher earns its cost against a rule based one, using transaction reconciliation as the test bed because ground truth can be constructed and the audit requirement forbids unexplainable matches.
- Three passes ordered by cost: a vectorised hash join, a tolerance pass over a date window with greedy closest assignment, and a lexical similarity pass on the remainder. Blocking by amount bucket and date window reduces a 9.8 billion pair comparison to a tractable candidate set.
- Result: 95% match rate, zero mismatches against a synthetic ground truth with a known answer key, 4.25 seconds at 100,000 against 98,000 records, and zero inference calls. It replaced an embedding based engine that was slower, more expensive and less accurate, and both remain in the repository so the comparison is reproducible.
- Every match records the pass and basis that produced it, since an unexplainable match is unusable in an audited setting.

`Python, pandas, FastAPI, PostgreSQL, Celery`

Verification state as a first class output in an applied LLM system Feb 2026 – present

DebugAI, a launched product used as an experimental surface.

- Production system that assembles context for a language model from a runtime stack trace: forced inclusion of traced files, then embedding retrieval over the repository for code the trace never names, under a managed token budget, with a bounded second retrieval pass when the model reports insufficient context.
- Each proposed fix carries an explicit verification state: mechanically checked and passed, checked and failed, or never checked. Unchecked is displayed as unchecked rather than being presented alongside verified results.
- Error recurrence is tracked by SHA-256 signature per project, and a human confirmed fix is promoted to lead subsequent answers for that signature. Memory is project scoped so a fix never crosses users.
- Model routing by estimated difficulty with a cross provider fallback that preserves the response shape, and a cache keyed on error text, project and framework checked before any inference call.
- This is where the retrieval failure observed during the 2025 RAG work was addressed directly: a system that cannot report its own uncertainty produces confident wrong answers at a steady rate.

`Python, FastAPI, TypeScript, PostgreSQL, pgvector, Redis, Anthropic and OpenAI APIs`

Retrieval augmented generation Jun – Jul 2025

Summer training programme, RemarkSkill Education with IIT Hyderabad.

- Built embedding retrieval by cosine similarity with document chunking and context injection over a document corpus.
- Found chunk boundary selection, not prompt design, dominated answer quality: a passage split mid argument answers half a question regardless of how the query is phrased.
- Observed that retrieval failure produced fluent, confident, entirely wrong answers with no distinguishing signal, which became the design driver for later work on verification state.

`Python, vector databases, embeddings, NLP`

Applied and engineering work

Telecare Nabha, Smart India Hackathon 2025 2025

- Telemedicine platform for a district where 31% of households have internet access and the civil hospital operates 11 doctors against 23 sanctioned posts for 173 villages. Dual channel by design, with an IVR path over a button phone alongside the app, offline first records, and live pharmacy stock.

Inkline: photographed handwriting to font outlines Aug 2026

- Homography based un-warp from printed fiducials, Moore neighbour boundary extraction, corner detection over an angular window, and Schneider least squares cubic fitting with Newton-Raphson reparameterisation, implemented from scratch rather than wrapped, with binary font table assembly.

SecAudit: dependency free static analysis Mar 2026

- 60+ rules across the OWASP Top 10 in a single standard library Python file, with the false positive tradeoff chosen explicitly toward missing a real issue over flagging a safe line.

Technical skills

Programming: Python (primary), TypeScript, JavaScript, C, SQL

Machine learning: PyTorch, TensorFlow, scikit-learn, OpenCV, convolutional fine tuning, embeddings and retrieval, multimodal fusion, error analysis, evaluation design

Signals and hardware: PPG and IMU processing, motion artifact rejection, power budgeting, ngspice, KiCad, FreeCAD, nRF52, ESP32, MEMS sensors

Systems: FastAPI, PostgreSQL, Redis, Celery, Docker, pgvector, distributed job design

Mathematics: Linear algebra, probability and statistics, optimisation, numerical methods, geometric algorithms

Currently learning: Uncertainty quantification, probabilistic machine learning, diffusion models, radiometric calibration

Recognition and external validation

- Second place, research paper competition, Techno Udbhav 2025 inter college technical festival, for OptiXon health sensing eyewear.
- Smart India Hackathon 2025, team Glitch Guardians, problem statement PS25018: telemedicine access for rural healthcare in Nabha, Punjab.
- OptiXon design and workflow reviewed across multiple sessions with M.Tech students at IIT Hyderabad and a working hardware professional, who confirmed feasibility and suggested changes. Every design, simulation and firmware decision is my own.
- Certificates: Generative AI with Machine Learning, Ethical Hacking, and Data Science workshops, all IIT Hyderabad.

How I work

- I build the deterministic baseline before the learned model, so the model's contribution is a measurable number rather than an assumption.
- I label every figure by how it was obtained: measured, derived, estimated, or paper only. Estimates stay labelled as estimates even when it makes the work sound smaller.
- I seek review early and in person. The OptiXon design went through students, a working professional and a faculty researcher before the fabrication pack was exported.
- I record failures with their causes. Four of fourteen projects are paused or retired, and each carries a written reason.

Availability and goals

Available for twelve or more weeks full time in summer 2027, willing to relocate. Near term goal is a research placement where I can work on signal recovery, efficient inference, or uncertainty in generated outputs under supervision. Longer term I am deciding between a PhD and industry research, and I would like to make that decision from inside a lab rather than from outside one. English fluent, Hindi native. References available on request.